# PROJECT-BASED LEARNING

## PHASE-II EVALUATION

### Smart Driver Drowsiness & Safety System

Team ID: DSCPP-III-2026-T234

**Department of Computer Science & Engineering**
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

# Team & Mentor Details

## Team Information

**Team ID:** DSCPP-III-2026-T234
**Semester:** 3rd
**Section:** H, E, MLA, ML3
**Domain:** *Computer Vision & Safety*

## Team Members

1. *Ridvik Garg – 2028039*
2. *Prabhav Gulyani – 2027975*
3. *Nandini Gupta – 2028777*
4. *Antriksh Gehlot – 2028648*

## Mentor

*Mr. Kamal Kumar Gola*

**Interactions completed:** 04

Brief Discussion: Introduce the team, project domain, mentor, and responsibilities of the team members.

# Phase-I Recap & Feedback Incorporated

## Phase-I Status

*Proposed a webcam system to track driver eyes and head position to prevent accidents from sleepy drivers.*

### Feedback Received

- *Use proper DSA instead of standard vectors*
- *Split project into smaller testable modules*
- *Avoid false beeps on normal eye blinks*

### Changes Incorporated

- *Added Queue, Stack, and Linked List classes*
- *Divided code into Vision, DSA, and DB modules*
- *Checked longer eye closure before alerting*

Brief Discussion: Explain the major feedback received during Phase-I and clearly demonstrate how it has been

# Problem Statement & Finalized Objectives

## Problem Statement

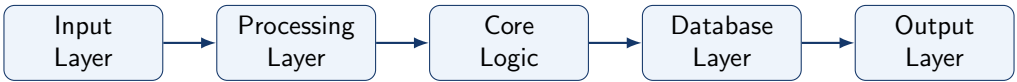
*Prevent accidents caused by sleepy drivers by tracking prolonged eye closure via webcam and giving step-by-step alerts.*

## Finalized Objectives

1. *Capture live webcam video with OpenCV to check eye closure and head position.*
2. *Use a Queue for incoming events, a Stack for recent alerts, and a Linked List for history.*
3. *Play step-by-step sound alerts, lower car music volume, and send emergency SMS.*
4. *Save all past safety events and logs inside a local SQLite database.*

Brief Discussion: Avoid repeating the complete Phase-I presentation. Focus on the finalized problem, scope and measurable objectives.

# Updated System Architecture / Workflow




## Major Components

- *OpenCV Camera & Face Tracking*
- *C++ Queue, Stack & Linked List Logic*
- *SQLite Database & Alert Triggers*

## Changes Since Phase-I

- *Added custom DSA classes for events*
- *Added music mute and SMS features*
- *Split into separate testable modules*

Brief Discussion: Present the updated architecture and explain how individual modules communicate with each other.

# Module Design & Functional Requirements

## Major Modules

1. *Webcam & OpenCV Vision Module*
2. *C++ DSA Event Handler Module*
3. *SQLite Database Storage Module*
4. *Sound, Music Mute & SMS Module*

## Functional Requirements

- *Track eyes and verify prolonged closure*
- *Push events to Queue and alerts to Stack*
- *Save event history in Linked List & DB*
- *Trigger beep, loud alarm, mute, and SMS*

### Module Dependency

*OpenCV spots eye closure and sends events to the C++ Queue, which updates the Stack, triggers alerts, and saves logs to SQLite.*

# Technology Stack & Development Methodology

## Technology Stack

- Programming: *C++*
- Database: *SQLite*
- Framework: *OpenCV*
- Development Tool: *VS Code & GCC*
- Version Control: *Git / GitHub*

## Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Brief Discussion: Explain why the selected technologies are appropriate and how the team is following the development methodology.

# Implementation of PBL Course Concepts

| Course | Concepts Actually Implemented | Project Module |
|---|---|---|
| **Data Structures in C** | *Queue (event order), Stack (recent alerts), Linked List (history)* | *DSA Event Module* |
| **OOPs with C++** | *Custom Classes, Objects, Encapsulation, Header Files* | *Core C++ Backend* |

**Important:** Mention only those concepts that have been **actually implemented** in the project.

Brief Discussion: Students should explain where and how each course concept is technically implemented.

# Development & Implementation Progress

| Module | Status | Work Completed |
|---|---|---|
| *C++ & DB Setup* | Completed | *Set up C++ repo, SQLite tables, and tested insert/read queries* |
| *Webcam Tracking* | Completed | *Tested webcam input and verified basic face tracking with OpenCV* |
| *DSA & EAR Logic* | In Progress | *Debugging Stack/Linked List classes and adjusting low-light EAR* |
| *Alerts & Merging* | Pending | *Sound alerts, music mute script, SMS API, and full module merge* |

**Overall Project Completion:** *55%*

Brief Discussion: Clearly differentiate between completed, in-progress and pending development. Do not present proposed work as completed work.

# Working Prototype / Demonstration

## Implemented Features

- *C++ build compiles cleanly*
- *Webcam live video capture*
- *Basic face & eye tracking*
- *SQLite insert & read queries*

### Prototype / Output

*Webcam face tracking & SQLite database test output ready*

Brief Discussion: Demonstrate the actual working implementation. Students should preferably provide a live demonstration in addition to screenshots wherever applicable.

# Testing, Results & Validation

| Test / Module | Expected Result | Observed Result | Status |
|---|---|---|---|
| *C++ & Webcam* | *Build files compile and camera runs* | *Compiles fine and captures video stably* | Pass |
| *DB Initialization* | *Insert and fetch safety records* | *Database is ready and queries work* | Pass |
| *DSA & Vision* | *No memory leaks; track in dark rooms* | *Fixing leaks and tuning low-light EAR* | *In Progress* |

## Results / Validation

*C++ build, webcam feed, and SQLite queries passed testing; eye tracking works in good light and is being tuned for dark rooms.*

Brief Discussion: Results should be supported by actual testing or experimental evidence rather than assumptions.

# GitHub Repository & Development Activity

## Repository Information

- GitHub Link: *github.com/ridvik74/Smart-Driver-Drowsiness-Safety-System/tree/main/DSA*
- README.md: *Updated*
- Contributors: *All 4 Members Added*
- Repository Status: *Public*
- Branches: *main*

## Development Activity

- *Custom DSA headers uploaded*
- *Camera processing scripts added*
- *Individual commits from own accounts*
- *SQLite database scripts uploaded*
- *Alert generation logic in progress*

**Requirement:** All team members must be contributors and must **commit individually using their own GitHub accounts**.

Brief Discussion: Show repository activity, README, contributor list and commit history during the presentation.

# Individual Contribution

| Member | Role | Technical Contribution | GitHub Commits |
|---|---|---|---|
| *Ridvik Garg* | *Team Lead* | *Updated DSA architecture; wrote event handling, Stack & Linked List* | *Active* |
| *Prabhav Gulyani* | *Backend & Vision* | *Wrote safety event logs; tested webcam & OpenCV face tracking* | *Active* |
| *Nandini Gupta* | *Vision & DB* | *Tested OpenCV face tracking; created SQLite DB and tested queries* | *Active* |
| *Antriksh Gehlot* | *DB & Alerts* | *Created SQLite DB and queries; working on sound, mute & SMS API* | *Active* |

## Individual Evaluation

Every member should be prepared to explain and demonstrate their own technical contribution to the project.

# Challenges, Remaining Work & Phase-III Roadmap

## Challenges Faced

- *Dark lighting and spectacles affect OpenCV*
- *Bugs while replacing vectors with custom DSA*
- *Crashes and CPU lag when linking modules*

## Solutions / Learning

- *Testing different EAR values for low light*
- *Module-by-module testing to fix memory leaks*

## Phase-III Roadmap

1. *Link eye-tracking with main event Queue*
2. *Finish debugging Stack and Linked List*
3. *Add sound alerts and music mute script*
4. *Set up SMS API for emergency texts*
5. *Merge all modules and do full testing*

# Expected Outcomes & References

## Expected Final Outcomes

- *Working driver safety prototype*
- *Real-time Queue, Stack & Linked List*
- *Step-by-step alarm and music mute*
- *Emergency SMS and SQLite logs*
- *Complete source code on GitHub*

## References


1. *Real-Time Eye Blink Detection (EAR Paper)*
2. *OpenCV Face Landmark Documentation*
3. *Data Structures in C/C++ Textbook*
4. *Official OpenCV & SQLite3 Docs*
5. *GitHub Repository: Smart-Driver-Safety*


**Thank You**